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ORIGINAL ARTICLE

Evaluating early vs. late static SUVR windows of [^{18}F]MK-6240 tau PET in Alzheimer disease: a head-to-head comparison study

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Abstract

Purpose [^{18}F]MK-6240 is a widely used second-generation tau PET tracer in Alzheimer disease (AD) and is under FDA review, making it important to refine practical static imaging windows for clinical use. Prior dynamic studies have supported late static windows (~90–110 min) because they improve agreement between standard uptake value ratios (SUVR) and kinetic models in high-binding regions, but extracerebral skull and meningeal uptake becomes more prominent at later times and may confound visual interpretation and SUVR quantification. In this study, we aim to determine whether an earlier static window (40–60 min) provides comparable discrimination of tau burden while reducing extracerebral spill-in and SUVR instability.

Methods In this retrospective within-subject study, 67 participants across the AD spectrum (normal controls (cognitively unimpaired) [NL], mild cognitive impairment [MCI], and AD) underwent [^{18}F]MK-6240 PET with early (40–60 min) and late (90–120 min) frames. SUVRs were computed in the cerebral cortex (CTX) and Braak ROIs using cerebellar cortex as reference. Early–late agreement was assessed using correlation, regression, and Kolmogorov–Smirnov (KS) tests. Diagnostic and tau-status classification performance was evaluated with ROC analyses. Skull/meningeal uptake was graded visually and quantified, and PET–MRI misregistration simulations assessed SUVR robustness. Time-resolved analyses (30–120 min) evaluated temporal changes in stability and discrimination.

Results Early and late SUVRs were highly correlated ($r \geq 0.97$; $p < 0.01$) and showed similar cohort-level distributions (CTX KS D = 0.060; $p > 0.99$). Discrimination was comparable for diagnosis and tau status (e.g., CTX AUC 0.88 vs. 0.85 for NL vs. MCI/AD). Late frames showed greater skull/meningeal uptake with increased spill-in and reference-region sensitivity; in tau-negative participants, skull SUVR exceeded entorhinal SUVR after ~60 min. Misregistration simulations showed greater variability for late versus early entorhinal SUVR ($\sigma = 0.071$ vs. 0.017). Time-resolved analyses showed that in this cohort, later acquisition did not materially improve diagnostic discrimination and was associated with more extracerebral contamination and instability.

Conclusion The 40–60-minute window provides discrimination comparable to 90–120 min while reducing extracerebral contamination and improving robustness, supporting shorter static [^{18}F]MK-6240 protocols for clinical and research applications.

Keywords [^{18}F]MK-6240 · tau PET · SUVR · Off-target binding · Alzheimer disease

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Abbreviations

AD	Alzheimer disease
MCI	mild cognitive impairment
NL	normal
SUVR	standard uptake value ratios
CTX	cerebral cortex
EC	entorhinal cortex
MTL	medial temporal lobe

Introduction

Alzheimer disease is characterized by amyloid-β (Aβ) plaques and tau neurofibrillary tangles (NFTs) [1, 2]. Because tau burden correlates more closely than Aβ with cognitive decline and disease severity [3], quantifying tau in vivo is essential for staging and tracking disease progression. Consequently, there have been sustained efforts in developing various tau-selective PET tracers [4–6], among which [^{18}F]MK-6240 has gained attention for its high affinity and favorable pharmacokinetics [7–9]. As this tracer undergoes FDA review, refining practical acquisition protocols is vital for clinical translation.

Current clinical practice largely favors late static imaging windows (~90–110/120 minutes postinjection) for SUVR [7, 10]. This convention is supported by kinetic studies showing that later frames yield SUVR measures that closely approximate reference-tissue distribution volume ratio (DVR) in high-binding regions (8, 17]; Test–retest work has further supported the stability of late-frame quantitative outcomes within dynamic-model comparisons (7]. Nevertheless, several practical and technical considerations warrant a reevaluation of this standard. First, [^{18}F]MK-6240 exhibits prominent off-target skull and meningeal uptake that increases over time, creating significant spill-in artifacts, particularly in the medial temporal lobe (MTL) and cerebellar tentorium [10–15]. These extracerebral signals may bias both cortical ROI values and the cerebellar reference region, and while methodological strategies like partial volume correction (PVC) can mitigate these effects, they introduce additional processing complexity and may leave residual contamination [16]. Second, prior work indicates that meningeal signals and certain small cortical ROIs may not uniformly reach equilibrium even in late frames, suggesting that delaying acquisition does not necessarily guarantee physiological stability across all regions [17]. Third, while much of the literature has emphasized SUVR-to-DVR agreement to justify late windows, the static window itself has not been systematically optimized for routine clinical conditions where patient comfort and motion-related noise are paramount.

Drawing from amyloid PET experience, imaging at earlier time points offers a pragmatic alternative. Modeling studies suggest that moderately earlier windows effectively capture the dominant specific-binding signal, with only minimal underestimation in regions of high tau burden [17]. Supporting this, truncated 40-minute dynamic protocols have shown distribution volume estimates within 10% error of 120-minute acquisitions, confirming that essential binding information resides in earlier frames [18]. Together, these observations motivate the question of whether an early static window can provide robust SUVRs while inherently reducing late-phase artifacts. To our knowledge, no definitive within-subject study has directly compared early and late static [^{18}F]MK-6240 windows across the AD spectrum.

To address this, we performed a within-subject, head-to-head comparison of early (40–60 min) versus late (90–120 min) [^{18}F]MK-6240 static imaging across the AD spectrum—cognitively normal older adults, individuals with amnestic MCI, and patients with AD dementia. We tested whether early-window SUVRs are comparable to late-window SUVRs for cortical tau quantification and whether earlier imaging reduces meningeal and skull spill-in artifacts. By quantifying regional SUVR agreement and the magnitude of extra-cerebral contamination, we evaluated (i) extracerebral uptake using visual grading and ROI-based measures, (ii) potential spill-in via proximity-based coupling and ROI erosion, and (iii) early–late comparability using correlation, distributional tests, and diagnostic classification performance.

Materials and methods

Study population

This retrospective study included 67 patients (26 men and 41 women; age > 55; mean age ± SD, 70.8 ± 8.4 y), comprising 42 NLs, 18 with MCI, and 7 with AD. Participants underwent standardized clinical assessment, 3T MRI, and [^{18}F]MK-6240 PET at the Weill Cornell Medicine Brain Health Imaging Institute. Executive and memory functions were assessed using the Clinical Dementia Rating (CDR) scale [19]. Subject diagnoses were made in conjunction with National Institute of Neurological and Communicative Disorders and Stroke/AD and Related Disorders Association criteria through consensus conferences involving cognitive neurologists, neuroradiologists, and neuropsychologists [20–22]. This study was approved by the WCM Institutional Review Board (IRB) and was conducted in compliance with the Health Insurance Portability and Accountability Act (HIPAA). Written informed consent was obtained from all participants.

MRI and PET acquisition

MRI was performed on a 3T Siemens Prisma scanner with a 64-channel head-neck coil, utilizing MPRAGE T1w and T2-SPACE sequence, which were identical to those described in our previous study [23]. PET scanning was conducted using a Siemens Biograph mCT–S (64) slice PET/CT. After a bolus injection of 152–219 MBq (mean ± SD, 193 ± 16) [^{18}F]MK-6240 (provided by Lantheus®), dynamic PET data were acquired in list mode for two sessions: 0–60 min and 90–120 min post-injection, with a 30-minute break between sessions. PET acquisition parameters and image processing followed previously published protocols [18]. Partial volume correction was not applied to avoid potential bias in window comparisons.

ROI parcellation

T1w MRI scans were processed using *FreeSurfer* version 7.1 with the *recon-all* command for ROI parcellation, incorporating T2w images to improve segmentation accuracy. ROIs were delineated using the FreeSurfer aparc+aseg parcellation (Desikan-Killiany atlas). Bilateral cerebellar cortex served as the reference region for calculating tau SUVR. Cerebral cortical ROIs and Braak stage composite regions were delineated for tau quantification, where Braak regions were defined based on the region labels listed in the UC Berkeley AV-1451 processing methods, and CTX was defined by aggregating labels 1000–2035 within the *FreeSurfer* look-up table. To minimize partial volume effects (PVE), all ROIs were eroded by 1 voxel, except the reference region, which was eroded by 2 voxels, using a 3-dimensional spherical kernel in MATLAB's *imerode* function. The cranial bone ROI was segmented using *SPM12* by applying a threshold of 0.4 to the skull probability map, excluding regions > 15 mm from brain tissue.

Data processing and analysis

SUVR quantification

SUVRs were calculated from summed images over specific time intervals: 30–40, 40–50, 50–60, 90–100, 100–110, 110–120, 40–60, and 90–120 min, using the cerebellar cortex as the reference region. ROI-wise SUVRs (including cerebral cortex, Braak stages I–VI, and skull) were extracted for quantitative data analysis, while whole-brain SUVR images for the 40–60 and 90–120 min intervals were generated for visual pathology assessment.

Visual assessment of tau status

Two radiologists (Y.L., G.C.; >15 years of experience) independently reviewed [^{18}F]MK-6240 PET SUVR 90–120 images co-registered with MRI. Scans were displayed in orthogonal views using an ACTC color scale (fixed SUVR range, 0–3), and visually assessed following published protocol [24]. Readers were blinded to all clinical information and quantitative SUVR metrics, with access only to PET and MRI images. Subjects were classified as **MTL positive** (MTL+; positive MTL uptake), **CTX positive** (CTX+; positive uptake in non-MTL neocortical regions), or **tau positive** (Tau+; positive uptake in any of the above regions).

Statistical analysis

Pairwise SUVR comparisons between the 40–60 min and 90–120 min windows were evaluated using Pearson correlation and simple linear regression (reported with 95% CI) across the CTX and Braaks ROIs. No additional covariates were included in these models, and no corrections for multiple comparisons were applied given the exploratory nature of the analysis.

To compare the distributions of 2 measurements, the **KS test** was used. This nonparametric test evaluates whether 2 samples are drawn from the same distribution by comparing their empirical cumulative distribution functions (ECDFs). The KS statistic (or D value) is defined as the maximum absolute difference between the 2 ECDFs, and it quantifies the degree of divergence between the distributions. A critical **value** ($D_{critical}$) corresponding to a significance level of $\alpha = 0.05$ was used to determine whether the null hypothesis—that both samples come from the same distribution—should be rejected [25].

The diagnostic performance of the models was evaluated using Receiver Operating Characteristic (ROC) curve analysis, with the Area Under the Curve (AUC) serving as the primary metric for discriminatory accuracy. To ensure the robustness of the results, 95% confidence intervals (CIs) and p-values for individual AUCs were derived using bootstrap resampling (e.g., $N = 5000$). For the comparison of performance between two ROC curves, the difference in AUCs was assessed using DeLong's test [26]. Statistical significance was defined as a two-tailed p-value < 0.05.

Inter-reader agreement for skull grades (0–3 scale was evaluated using weighted Cohen's kappa with quadratic weights: $w = 1 - (grade\,difference/3)^2$ [27]. All these statistical analyses were performed using RStudio (v2024.09.1; Posit) and MATLAB (v2025a; MathWorks).

Results

Participants and imaging overview

The analysis included 67 participants (mean age ± SD, 70.8 ± 8.4; 41 women), comprising 42 NL, 18 MCI, and 7 AD. Demographic data are summarized in Table 1. Age and sex distribution showed no significant differences between the NL and MCI/AD groups ($p > 0.05$).

Table 1 The demographic and diagnostic information of study participants

Item	NL	MCI/AD
Total	42	25 (18/7)
Sex: Male (%)*	13 (31%)	13 (54%)
Mean Age (SD)**	71.35 (7.55)	69.77 (9.73)
CDR	=0	>=0.5
Tau Positive	13 (31%)	22 (88%)

*A chi-squared test of independence indicated no significant association between sex and diagnostic group (p=0.09)

**A two-tailed t-test showed no significant difference in age between the diagnostic groups (p=0.49)

Visual and quantitative assessment of extracerebral (skull/meningeal) contamination

Visual assessment

Visual examination of the SUVR images revealed more prominent skull and meningeal [18 F]MK-6240 uptake in 90–120 min ($SUVR_{90-120}$) images compared with 40–60 min ($SUVR_{40-60}$) images (Fig. 1A). These extracerebral signals were observed in both tau-positive and tau-negative participants (Fig. 1B). To systematically evaluate this, we employed a visual grading system (grades 0–3) incorporating signal intensity and spatial pattern (details in Appendix A), where grade 0 indicated minimal signal and grade 3 indicated strong uptake. This grading system demonstrated consistency with quantitative measurements (Fig. 2; Table 2) and high inter-reader reliability (weighted Cohen κ=0.82; > 0.80 indicates excellent agreement). Grade≥2 skull uptake was present in 42% (28 of 67) of $SUVR_{90-120}$ images, whereas 0% (0 of 67) of $SUVR_{40-60}$ images exhibited this level of uptake (Table 3).

Extracerebral spill-in quantitative analysis

In the MTL negative group, skull SUVR exceeded the entorhinal cortex (EC) SUVR after approximately 60 min (Fig. 3A). Coupling analysis between EC and adjacent skull SUVR (within 3 mm) showed a correlation of r_(early) = 0.31 for $SUVR_{40-60}$, which increased to r_(late) = 0.42 for $SUVR_{90-120}$ (p = 0.07 for r_(late) > r_(early) via bootstrap test). To verify if this increased late-window correlation was driven by spill-in, we performed one-voxel erosion of the EC mask. Following erosion, these correlations decreased to r_(early) = 0.23 and r_(late) = 0.38, respectively.

To evaluate if meningeal spill-in similarly affected the reference region, we compared eroded and non-eroded cerebellar cortex masks. Omitting two-voxel erosion resulted in an average 4% overestimation of cerebellar SUV at 90–120 min due to meningeal signal compared with a bias of <2.1% at 40–60 min (Fig. 3B). The SUV differences between uneroded and eroded cerebellar ROIs were larger and more variable in the late window, particularly in tau-positive participants.

Comparison of $SUVR_{40-60}$ and $SUVR_{90-120}$

Agreement between SUVR40–60 and SUVR90–120

Pearson correlation analysis demonstrated high consistency between the two windows across regions with r > 0.99 in the CTX (Fig. 4A) and r = 0.98 in the EC (Braak stage I), which represents one of the earliest regions affected by tau pathology (Fig. 4B; both p < 0.001). Additional Braak ROIs showed similarly high correlations (all r > 0.97, p < 0.001;

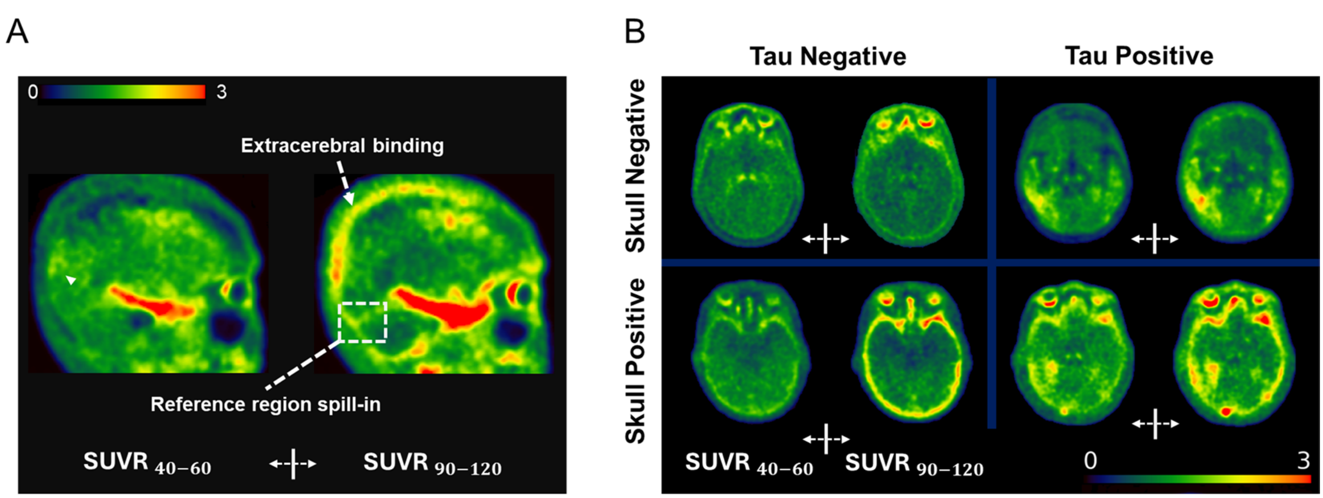

Fig. 1 (**A**) Example subject showing early (40–60 min) and late (90–120 min) SUVR images. The late window shows stronger meningeal uptake (white box) and skull signal (white arrow). A focal cortical tau signal (white triangle) in the inferior parietal cortex is partially obscured by skull spill-in in the late window. (**B**) Representative early and late SUVR images from four subjects, stratified by tau status (columns) and presence/absence of skull/meningeal binding (rows). Color scale: 0–3 SUVR

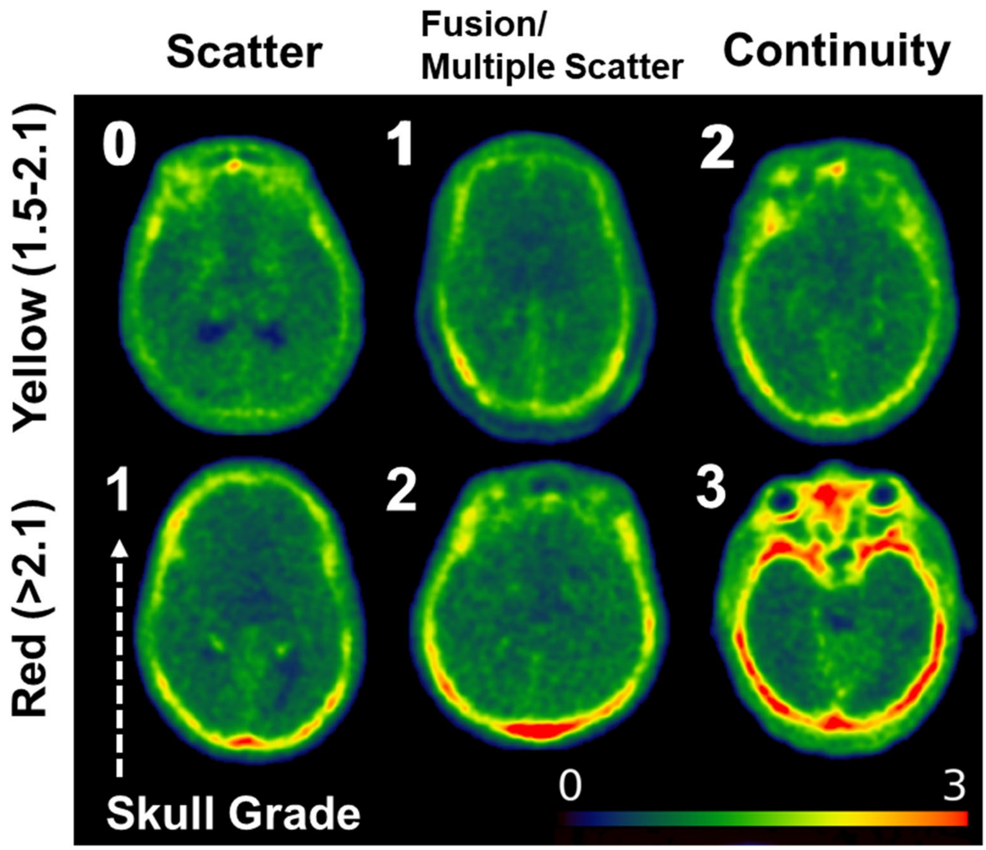

Fig. 2 Skull binding is classified by pattern (columns) and intensity (rows). Patterns are categorized as: scatter (localized focal points appearing as isolated bright spots across any anatomical axis), fusion (merged signals forming distinct patches or large arcs visible across multiple slices), and continuity (near-complete or full circumferential binding persistent across the majority of slices). Intensity is categorized by SUVR thresholds: yellow (SUVR 1.5–2.1) and red (SUVR > 2.1). Each image is assigned a grade (0–3, indicated in the top-left) based on the combined spatial extent and intensity, as defined in Table 2

Table 2 The skull grading system by visual pattern

Intensity/Patterns	Scatter	Fusion/ Multiple scatter	Continuity
Yellow	0	1	2
Red	1	2	3

Grade	Explain
0	Low intensity or at most Scatter with yellow intensity
1	One of the following occurs: 1. Fusion with yellow intensity 2. Scatter with red intensity
2	One of the following occurs: 1. Continuity with yellow intensity 2. Fusion with red intensity
3	Continuity with red intensity

Table 3 Cross-validation between skull grade and tau status (% relative to total 67 subjects)

Skull Grade in Early Time Frame					
Grade	0	1	2	3	Skull+ Subjects (Grade 1–3) / Row Total
Tau+	28(42%)	7(10%)	0(0%)	0(0%)	7/35
Tau-	23(34%)	9(13%)	0(0%)	0(0%)	9/32
Total	51(76%)	16(24%)	0(0%)	0(0%)	
Skull Grade in Late Time Frame					
Grade	0	1	2	3	Skull+ Subjects (Grade 1–3) / Row Total
Tau+	10(15%)	12(18%)	10(15%)	3(4%)	25/35
Tau-	3(21%)	14(21%)	9(13%)	6(9%)	29/32
Total	13(19%)	26(39%)	19(28%)	9(13%)	

Fig. A2). Subgroup analyses were consistent within CTX (NL: $r = 0.91$, $p < 0.001$; MCI/AD: $r = 1.00$, $p < 0.001$) and EC (NL: $r = 0.88$, $p < 0.001$; MCI/AD: $r = 0.98$, $p < 0.001$) (Fig. A3). However, despite these high correlations, Bland-Altman analysis indicated that the two windows are not strictly interchangeable due to significant proportional biases (ratio outside the LoA; CTX 7.5%, EC 10.4%) (Fig. A4).

To further evaluate if the data characteristics—such as the spread and dynamic range of values across subjects—were comparable between windows, we performed two-sample KS tests on z-standardized SUVRs. This cohort-level analysis showed no systematic shifts or differences in data distribution between the early and late windows for either the CTX (D = 0.060, $p > 0.99$; Fig. 4C) or EC (D = 0.075, $p = 0.99$; Fig. 4D), with both test statistics falling below the critical threshold ($D_{critical} = 0.235$). Similar results were observed across other Braak ROIs (Fig. A2), indicating that the overall cohort-level distributional shapes are consistent between the early and late windows, with no evidence of systematic distributional bias.

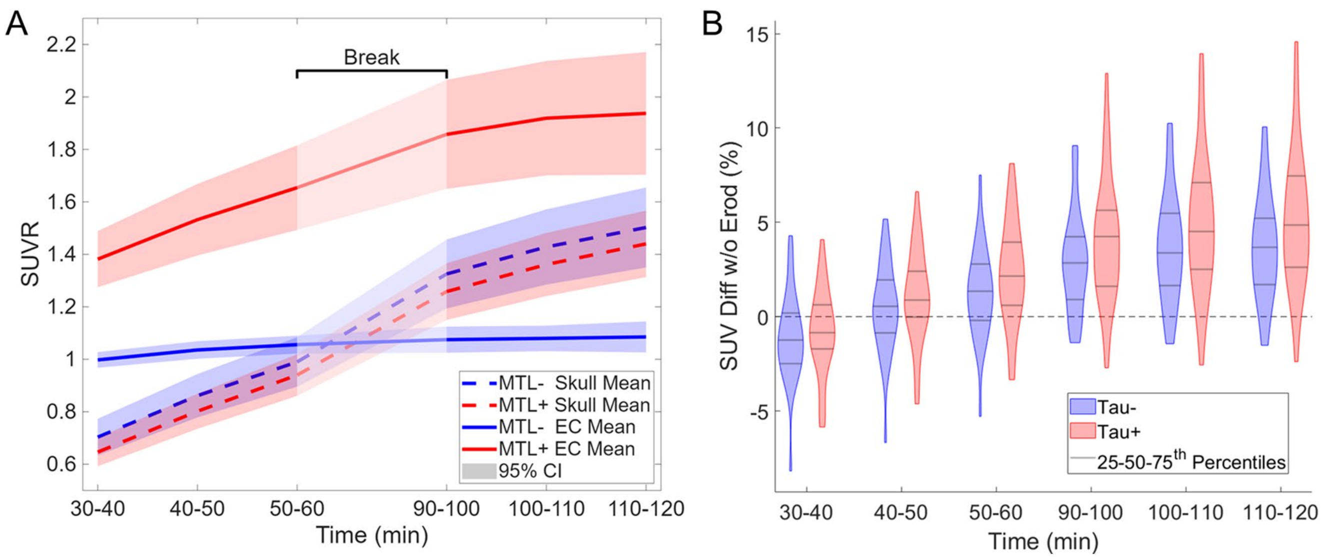

Fig. 3 (**A**) Group-average SUVR trajectories for the skull and entorhinal cortex (EC) from 30–120 min post-injection (10-min bins). There is a break between 60 and 90 min. The curves show group-wise means with shaded 95% confidence intervals. (**B**) Violin plots of SUV Diff w/o Erod (%), defined as the percentage SUV difference between uneroded and eroded cerebellar cortex (SUV_no-erod−SUV_erod)/SUV_erod×100 (%), shown across time windows for Tau− (blue) and Tau+ (red) groups

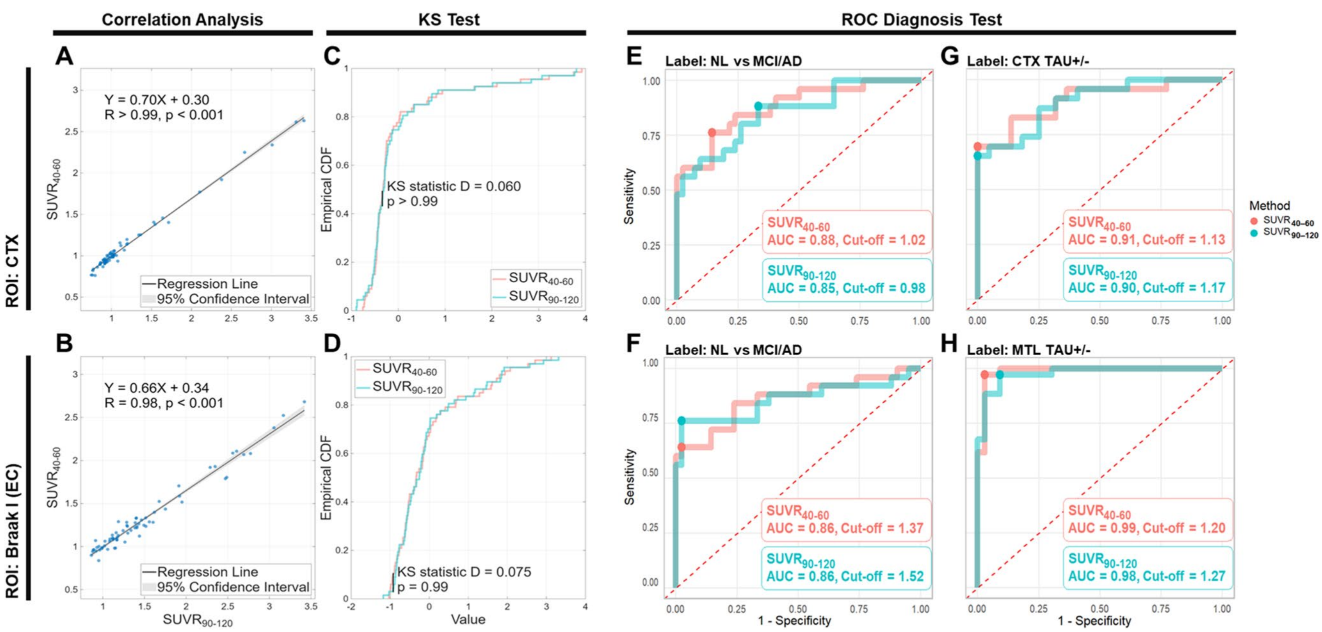

Fig. 4 (**A**–**B**) Correlation between $SUVR_{40–60}$ and $SUVR_{90–120}$ in (**A**) CTX and (**B**) Braak I (EC). (**C**–**D**) Two-sample KS tests comparing z-standardized SUVR distributions between windows in (**C**) CTX and (**D**) Braak I. (**E**–**F**) ROC curves for CTX and Braak I SUVRs distinguishing NL vs. MCI/AD (bold dot: optimal cut-off maximizing sensitivity+specificity). (**G**–**H**) ROC curves for CTX and Braak I SUVRs predicting tau positivity in the corresponding regions

Diagnostic performance

Both measures showed strong discrimination between NL and MCI/AD groups by two-tailed *t*-test in CTX ($SUVR_{40–60}$: t=5.39, $p<0.01$, $SUVR_{90–120}$: t=5.18, $p<0.01$) and EC ($SUVR_{40–60}$: t=6.88, $p<0.01$, $SUVR_{90–120}$: t=6.97, $p<0.01$). To further assess the diagnostic performance of $SUVR_{40–60}$ and $SUVR_{90–120}$, we performed ROC analyses in the CTX and EC to differentiate MCI/AD patients from NL participants. In the CTX (Fig. 4E), $SUVR_{40–60}$ achieved an AUC of 0.88 (95%CI, 0.78, 0.96; $p<0.01$) with an optimal cut-off value of 1.02 to separate MCI/AD from NL in CTX (sensitivity = 0.76 [19/25], specificity = 0.86 [36/42]). In comparison, $SUVR_{90–120}$ yielded an AUC of 0.85 (95%CI, 0.74, 0.94; $p < 0.01$) with a cut-off of 0.98 (sensitivity = 0.88 [22/25], specificity = 0.67 [28/42]), no significant difference between AUCs (Delong's $p = 0.26$). For the EC (Fig. 4F), $SUVR_{40–60}$ yielded an AUC of 0.86 (95%CI, 0.74, 0.96; $p < 0.01$) with a cut-off of 1.37 (sensitivity = 0.64 [16/25], specificity = 0.98 [41/42]), while $SUVR_{90–120}$ showed a comparable AUC of 0.86 (95%CI, 0.73, 0.96; $p < 0.01$) with a cut-off of 1.52 (sensitivity = 0.76 [19/25], specificity = 0.98 [41/42]), no significant difference between AUCs (Delong's $p = 0.92$).

To evaluate how SUVR measures from early and late scan windows correspond to clinical visual assessments of tau PET scans, we analyzed $SUVR_{40–60}$ and $SUVR_{90–120}$ in the CTX and MTL separately. For predicting tau in the CTX based on clinician reads (Fig. 4G), $SUVR_{40–60}$ achieved an AUC of 0.91 (95%CI, 0.81, 0.98; $p < 0.01$) with an optimal cut-off of 1.13 (sensitivity = 0.70 [16/23], specificity = 1.00 [44/44]), while $SUVR_{90–120}$ yielded a comparable AUC of 0.90 (95%CI, 0.81, 0.97; $p < 0.01$) with a cut-off of 1.17 (sensitivity = 0.65 [15/23], specificity = 1.00 [44/44]), no significant difference between AUCs (Delong's $p = 0.75$). Similar ROC analyses were conducted using $SUVR_{40–60}$ and $SUVR_{90–120}$ in the EC to predict tau positivity in the MTL as determined by clinician reads (Fig. 4H). For predicting MTL tau positivity, $SUVR_{40–60}$ yielded an AUC of 0.99 (95%CI, 0.96, 1.00; $p < 0.01$) with an optimal cut-off of 1.20 (sensitivity = 0.97 [33/34], specificity = 0.97 [32/33]), while $SUVR_{90–120}$ showed an AUC of 0.98 (95%CI, 0.94, 1.00; $p < 0.01$) with a cut-off of 1.27 (sensitivity = 0.97 [33/34], specificity = 0.91 [30/33]), no significant difference between AUCs (Delong's $p = 0.57$).

Comparison of reliability and diagnostic performance

SUVR robustness analysis

Visual assessment revealed prominent extracerebral signal around the cerebellum, particularly along the tentorium cerebelli, which was more pronounced in late frames than in early frames. While eroding the reference region may partially reduce spill-in, high surrounding signals can still introduce instability in SUVR calculations. To evaluate this, we conducted a misregistration sensitivity analysis by randomly shifting the EC and reference masks together (±3° rotation, ±1 mm translation; Fig. 5A) over 1,000 trials to recalculate EC SUVR. The EC was selected due to its early involvement in tau pathology and susceptibility to spill-in from its proximity to the skull base. The resulting SUVR distributions (Fig. 5B) showed that SUVR90–120 demonstrated a substantially wider spread than SUVR40–60 (σ_(early) = 0.017, σ_(late) = 0.071), with an SD ratio of 4.18.

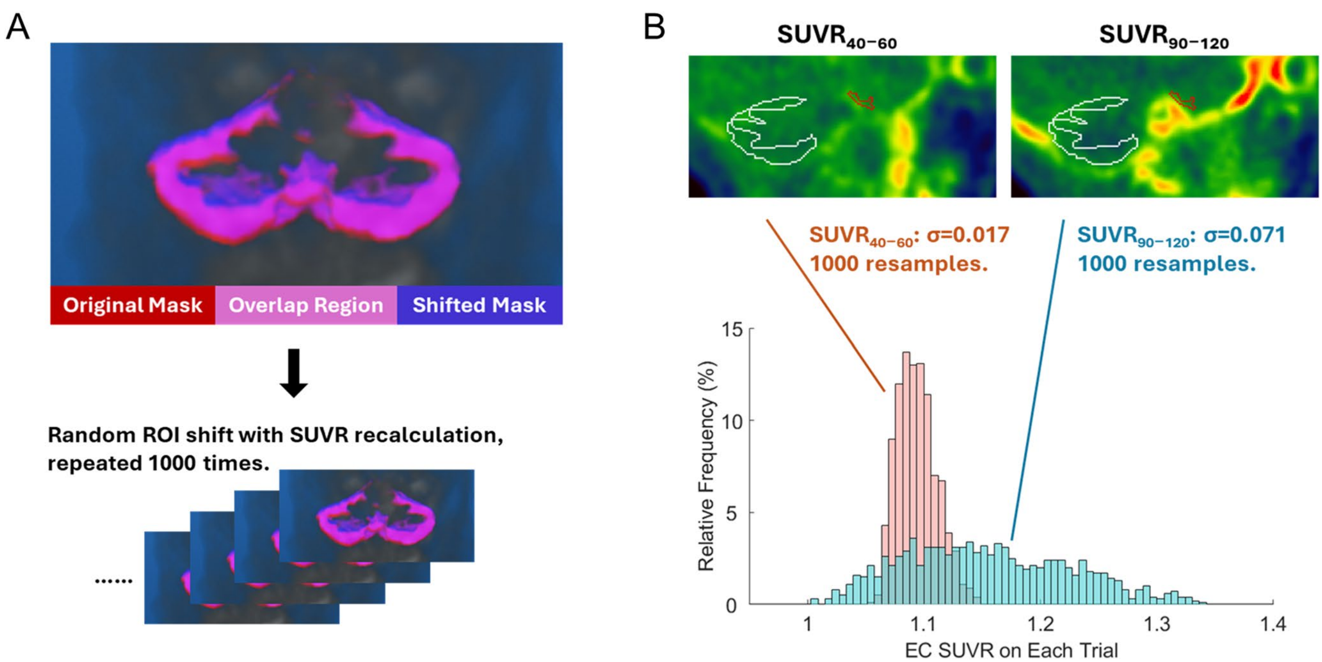

Fig. 5 Illustration of the SUVR instability test. (**A**) Top: example of a single random ROI shift, with a close-up view of the cerebellar cortex (original mask in red, shifted mask in blue, overlap in purple). In the same trial, the corresponding shift was also applied to the entorhinal cortex (EC) to calculate EC SUVR. Bottom: this shift-and-recalculation procedure was repeated 1,000 times. (**B**) For each shifted mask, EC $SUVR_{40-60}$ and $SUVR_{90-120}$ were recomputed (EC mask in red, cerebellar cortex reference in white). The histograms show the distribution of SUVR values across all 1,000 shifts and the corresponding standard deviation (σ), demonstrating greater variability in the late frame

Time-resolved analysis of diagnostic performance (30–120 min)

To examine how diagnostic performance evolves over time, we computed SUVR in 10-minute windows from 30 to 120 min in the CTX. While mean SUVR increased in tau-positive subjects, the separation between groups remained stable across all intervals (Fig. 6, top). Two-tailed two-sample t-tests confirmed consistent group differences among time (middle row), and ROC analyses demonstrated that the ability to predict tau positivity (AUC) remained high throughout the scan duration (bottom row). Similar time-resolved analysis in the MTL yielded significant t-values ($t > 7$) from 30 to 120 min, though a slight decrease in AUC was observed in the latest time frames (Fig. A5).

Discussion

In this study, we systematically re-evaluated the imaging window for [^{18}F]MK-6240 by addressing the persistent challenge of late-frame extracerebral contamination. Our investigation followed three primary axes: **first**, we documented that skull and meningeal binding are not driven by underlying tau pathology and confound late-frame (90–120 min) visual and quantitative assessments. **Second**, we demonstrated high agreement in SUVR and diagnostic performance between early and late windows. **Finally**, through misregistration simulations and time-resolved analyses, we examined the reliability of SUVR and its capacity for pathology discrimination.

Skull and meningeal binding & spill-in to cortical SUVR

Visual validation

A central practical finding of this study is that skull/meningeal uptake becomes a frequent and visually prominent feature in the late 90–120 min window, while it is far less apparent in the 40–60 min window. Using a validated 4-level visual grading scheme ($\kappa = 0.82$) that tracked quantitative skull SUVR (Appendix A), we found that clinically meaningful extracerebral uptake (grade ≥ 2)

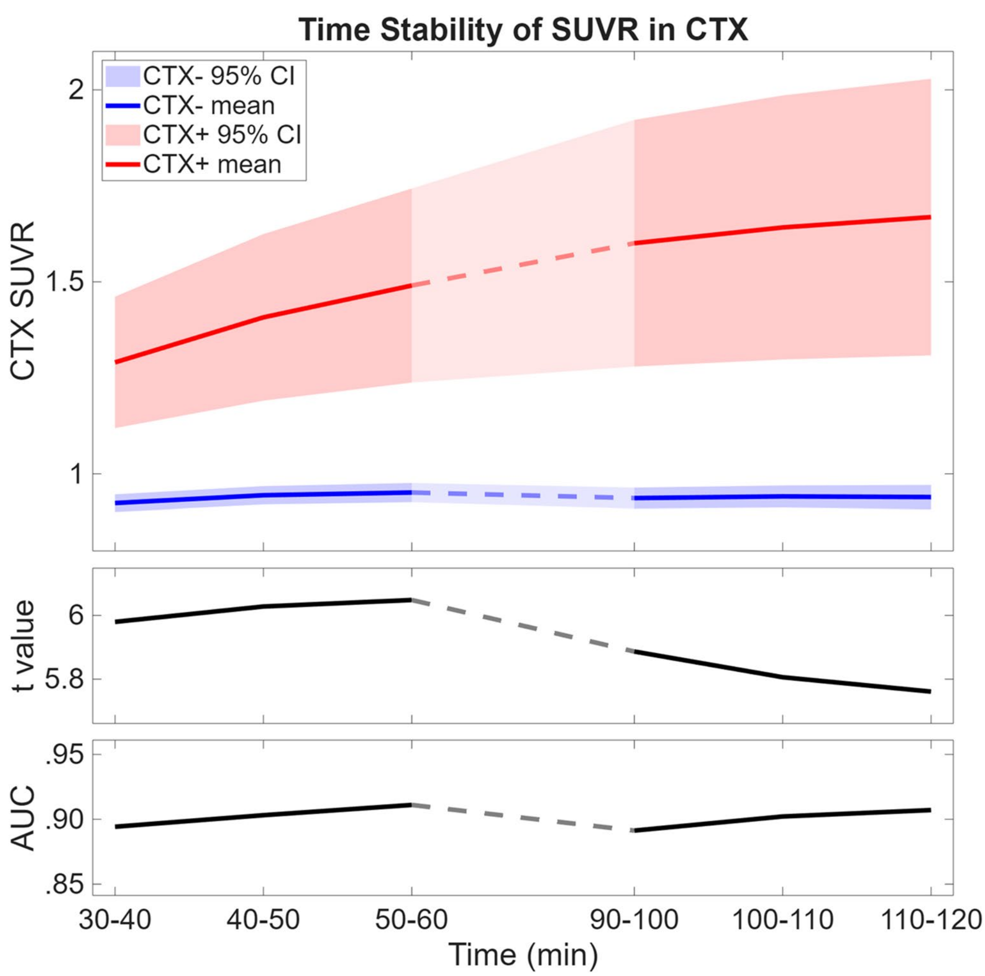

Fig. 6 Top: Mean SUVR and 95% confidence intervals over time in CTX, stratified by tau status (positive vs. negative). Middle: Corresponding t-values from two-sample two-tailed t-tests at each time window. Bottom: AUC values quantifying the ability of SUVR to discriminate tau positivity at each time point

was present in 42% of late-window scans and in none of the early-window scans. Importantly, this extracerebral signal was observed regardless of tau status (Fig. 1B), providing no evidence that it reflects underlying tau pathology, which agrees with previous findings [12, 28]. Although some studies report weak age associations or modest increases over scan time, the extracerebral signal shows high inter-individual variability and poorer test–retest reliability than cortical regions, supporting its non-specific nature [11, 15, 29, 30]. All those findings reinforce extracerebral binding as an off-target phenomenon that can complicate late-frame reads. The biological basis of meningeal/skull uptake remains under study. Prior work implicates binding to melanin-containing leptomeningeal cells [17, 31]. The temporal increase we observe suggests additional contributions from circulating radiometabolites that, while typically not crossing the BBB [8, 32], may accumulate in the meningeal vasculature lacking a classical BBB. Minor defluorination could contribute, but is unlikely to fully explain the late-phase rise in meningeal signal.

Spill-in to cortex

Consistent with these concerns, skull/meningeal activity often occurs along cerebral surfaces and can contaminate adjacent superficial cortex. Specially in the MTL, where extracerebral cortex separation is minimal, spill-in can mimic focal cortical binding or obscure true cortical signal, complicating visual interpretation and lowering reader confidence. Beyond visual confounding, our analyses suggest a measurable impact on quantitative estimates in regions adjacent to the skull base. In tau-negative participants, skull/meningeal SUVR exceeded entorhinal cortex SUVR after ~60 min, increasing susceptibility of MTL estimates to spill-in from adjacent extracerebral uptake. Consistent with this, coupling between the EC/MTL ROI and an adjacent skull ROI (within 3 mm) increased from the early to late window ($r_{(early)}=0.31$ vs. $r_{(late)}=0.42$; bootstrap $p=0.07$). Eroding the EC/MTL ROI reduced coupling in both windows ($r_{(early)}=0.23$; $r_{(late)}=0.38$), yet the late association remained stronger, supporting greater susceptibility to spill-in at 90–120 min even after conservative ROI refinement.

Spill-in to reference-region

Reference-region contamination represents an additional late-window vulnerability. Extracerebral uptake along the tentorium frequently contaminated the cerebellar cortex, the standard SUVR reference for [^{18}F]MK-6240. In sensitivity analyses, omitting cerebellar ROI erosion increased the measured cerebellar signal by >4% on average at 90–120 min (vs. <2.1% at 40–60 min) and increased inter-subject variability, particularly among Tau+ participants. These findings also support our results that late-window SUVR is more sensitive to reference-mask definition and to extracerebral signal near the reference region—both of which can propagate instability into SUVR estimates.

Taken together, skull/meningeal uptake is not only a common visual confound in late [^{18}F]MK6240 frames but is also linked to spill-in and reference contamination, underscoring the need for heightened caution when interpreting cortical foci adjacent to intense extracerebral signal on late window images.

Agreement, calibration and diagnostic performance

We systematically compared SUVR measurements from the 40–60 min and 90–120 min post-injection windows for [^{18}F]MK-6240 to assess whether earlier imaging can serve as a practical alternative for quantifying tau burden while reducing late-phase extracerebral artifacts. Across the cerebral cortex and Braak-stage ROIs, $SUVR_{40-60}$ and $SUVR_{90-120}$ showed strong concordance, with high regional correlations ($r>0.97$ for all ROIs). Notably, regression slopes deviated from unity (CTX: 0.70; EC: 0.66), so the two windows are not numerically interchangeable without calibration. However, at the cohort level, normalized distributions (z-scored) did not differ by scan windows (KS $p>0.99$). This consistency in overall data shape indicates that adopting an earlier window does not introduce structural distributional biases, thereby providing a well-aligned foundation for threshold-based categorization.

Reflecting this distributional alignment, $SUVR_{40-60}$ closely preserved clinically relevant contrasts despite these scale differences. Group comparisons showed robust separation between NL and MCI/AD in both the CTX and EC across both windows (all $p<0.001$), and ROC analyses demonstrated similar diagnostic performance between early and late imaging (CTX: 0.88 vs. 0.85; EC: 0.86 vs. 0.86, both Delong's $p>0.05$) with window-specific cutoffs (CTX 1.02 vs. 0.98; EC 1.37 vs. 1.52). The thresholds are consistent with previous reports [33]. Importantly, early- and late-window SUVRs also corresponded similarly to clinician visual reads: for predicting cortical tau positivity, $SUVR_{40-60}$ and $SUVR_{90-120}$ achieved comparable ROCs, and for predicting MTL tau positivity, both windows performed near ceiling with only minimal differences. Together, these findings support early-window SUVR as a clinically usable measure of tau burden, while emphasizing that direct numerical interchangeability with late-window SUVR requires calibration.

Robustness to processing variability

Although early and late SUVRs are highly correlated, practical use also depends on robustness to processing variability. Because extracerebral uptake increases over time, late frame SUVRs are expected to be more vulnerable to spill-in–related sensitivity to small PET–MRI misalignments commonly encountered in image processing. To quantify this, we performed a misregistration sensitivity analysis by jointly perturbing the EC and reference masks with realistic rotations (±3°) and translations (±1 mm) over 1,000 trials. Late window EC SUVR exhibited substantially greater dispersion than early window EC SUVR (σ_(early) = 0.017 vs. σ_(late) = 0.071), corresponding to a ~17fold increase in variance, indicating that $SUVR_{90-120}$ is more susceptible to routine processing variability, whereas $SUVR_{40-60}$ is comparatively robust under these simulated perturbation conditions.

Time-resolved diagnostic performance

Using dynamic PET data summarized in 10-minute windows from 30 to 120 min (CTX), we observed that diagnostic performance for tau positivity was already high by ~60 min, and extending acquisition beyond this time did not improve classification in the cortex (Fig. 6). Although between-group mean SUVR differences increased slightly later in the scan, the accompanying increase in variability (widening confidence intervals and reduced t-statistics) indicates diminishing returns for late acquisition. In line with prior reports that meningeal SUVR and small cortical ROIs may not uniformly reach equilibrium even in late frames [17], our emphasis here is not on achieving complete equilibrium of all extracerebral signals, but on identifying a pragmatic window where SUVRs already separate clinical groups reliably while minimizing variance-inflating, extracerebral influences. Collectively, these observations support 40–60 min as a practical static window that balances stability and diagnostic separability.

Limitations

This study has several limitations. First, the sample size, particularly for the AD dementia subgroup, was modest, which may limit the precision of subgroup-specific estimates and optimal cutoffs. In addition, pooling MCI and AD in some analyses may obscure potential stage-specific differences in tracer behavior, particularly in subjects with higher cortical binding.

Second, our conclusions are based on SUVR rather than full kinetic modeling. Although SUVR is widely used clinically, it remains sensitive to reference-region definition and acquisition timing. While the earlier 40–60 min window showed comparable classification performance, we did not perform a direct comparison with kinetic modeling in this dataset. Therefore, it remains uncertain whether the earlier window preserves quantitative fidelity to underlying tracer binding. Furthermore, because this study was not designed as a formal equivalence trial, the absence of significant AUC differences does not establish diagnostic equivalence, and caution is warranted when extending these findings. In addition, although this study primarily focused on group-level comparisons, preliminary region-wise analyses are provided in Appendix F for reference.

Finally, the present single-center sample was not sufficient to establish whether these findings will generalize across scanners, processing pipelines, and patient populations. In particular, extracerebral uptake patterns and processing choices—such as erosion strategies, reference-region definitions, and PET-MRI registration—may vary across sites. Crucially, while our misregistration simulation involved simplified, uniform shifts, real-world errors are often spatially heterogeneous or non-linear. This analysis should therefore be interpreted as a controlled stress test of window stability rather than a realistic estimate of clinical error propagation. These factors underscore the need for larger multi-center validation and future harmonization across acquisition windows.

Harmonization/translational implications

From a translational perspective, the potential adoption of a 40–60 min protocol will depend not only on its diagnostic performance, but also on its compatibility with the large body of existing [^{18}F]MK-6240 literature based on later acquisition windows. Our findings suggest that early-window SUVR values are promising for classification-oriented use, but they should not be considered directly interchangeable with conventional 90–120 min values. Practical implementation will therefore require harmonization strategies that map early-window measurements onto the established late-window framework, including empirical calibration equations, window-specific thresholds, and validation in larger multi-site datasets. A common-scale harmonization approach analogous to Centiloid may ultimately help improve comparability across acquisition windows, sites, and legacy datasets. Our results further suggest that such calibration may need to account for nonlinearity and for the influence of extracerebral off-target binding on cortical SUVR. In this context, the skull grading framework proposed in this study may provide a useful basis for future normalization efforts.

Conclusion

Taken together, the 40–60 min window offers a pragmatic alternative to 90–120 min for [^{18}F]MK-6240: it provides similar discriminative performance, reduces susceptibility to meningeal spill-in (including near the cerebellar reference), and improves robustness to routine processing variability, while also shortening scan time. These findings support earlier static imaging as a potentially advantageous option for clinical and research settings that rely on SUVR-based classification and visual interpretation.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval The study was approved by the IRB and was conducted in compliance with HIPAA.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent to publish The authors affirm that human research participants provided informed consent for publication.

Competing interests The authors declare no conflicts of interest and no industry support that could directly or indirectly influence this work.

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